

EXOhSPEC Stage-2 Thermal–Mechanical Stabilisation

Hybrid, TEC-only, AO-only, and Open-loop Baselines (V11–V14), and
Phase-Correlation Frame Registration

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July 13, 2026

Abstract

I compared four feedback configurations of the EXOhSPEC Stage-2 spectrograph stabilisation controller: the hybrid TEC+AO controller (V11, 31.4 h), a TEC-only variant with AO disabled (V12, 2.95 h), an AO-only variant with TEC frozen after warmup (V13, 4.06 h), and an extended hybrid run followed by a deliberate open-loop test (V14). Hybrid, AO-only, and V14’s clean hybrid segment all achieve comparable feedback-phase ΔY RMS (0.121–0.152 px, $\geq 99.9\%$ of frames within ± 0.5 px); TEC-only is the weakest closed-loop configuration (0.316 px, 88.8%), consistent with its slower thermal bandwidth and the short run length rather than a control failure. A detailed re-examination of the V14 CSV, cross-checked against lab notes and raw timestamps, revealed that its feedback phase actually contains three distinct control regimes rather than one: a clean, uninterrupted 30 h hybrid run (dY RMS 0.128 px), an unplanned ~ 9 h restart segment in which the OPL model was inactive and AO alone held ΔY to 0.086 px while ΔX drifted to a 0.42 px offset, and the true open-loop segment (dY RMS 1.357 px). The corrected open-loop-to-clean-hybrid RMS ratio is $10.6\times$. Separately, I diagnosed and fixed a phase-correlation image-registration pipeline for frame-to-frame drift measurement: switching from phase-only to standard cross-correlation normalisation, with a tuned ~ 100 – 200 px ROI, raised agreement with the live centroid tracker from 0.19–0.53 to 0.995–0.998 correlation.

Keywords: EXOhSPEC, thermal-mechanical stabilisation, closed-loop control, TEC, adaptive optics, centroid tracking, phase correlation, radial velocity instrumentation

1 Introduction

EXOhSPEC’s Stage-2 controller holds the spectrograph’s image centroid stable against thermal and mechanical drift using two actuators: a thermoelectric cooler (TEC) providing slow, large-range correction, and a tip-tilt adaptive-optics (AO) mirror providing fast, small-range correction. V11 is the last known-good hybrid (TEC+AO) baseline. Three further configurations isolate each actuator’s contribution and measure the uncorrected baseline directly:

- **V12** – AO disabled, TEC-only.
- **V13** – TEC frozen after warmup, AO-only.
- **V14** – Hybrid (TEC+AO) as V11, run longer, followed by a genuine open-loop segment (TEC, external fan, and air conditioning physically switched off). V14 also adds a fifth OPL-model predictor, outdoor temperature ($g_{T,\text{out}}$).

Separately, I developed and validated a phase-correlation-based frame-registration method as an independent cross-check on the live centroid tracker, and diagnosed why the first implementation of that method failed. Both workstreams are reported here.

Table 1 previews the headline result, colour-coded for quick reading: green cells indicate strong, consistent performance; yellow indicates a run with a real but explainable limitation; red flags the open-loop baseline, included deliberately as the reference for "no control at all."

Table 1: Headline comparison, colour-coded (green = strong & consistent, yellow = limited but explainable, red = uncontrolled baseline).

Run	Config	ΔY RMS (px)	Within ± 0.5 px (%)
V11	Hybrid (TEC+AO)	0.152	100.0
V12	TEC-only	0.316	88.8
V13	AO-only	0.138	100.0
V14 Seg A	Hybrid (TEC+AO)	0.128	100.0
V14 Seg B	Restart fallback (AO-only, no OPL model)	0.086	100.0
V14 Seg C	Open-loop (no control)	1.357	59.4

2 Method

All four controller runs share the same architecture: a 90-minute warmup phase (model frozen to seeded coefficients) followed by a feedback phase in which a ridge-regression pixel model and an optical-path-length (OPL) model drive a gain-scheduled PI controller on the TEC setpoint, with AO providing sub-pixel dithering once the pixel error is small and stable.

V14 timeline reconstruction – what the raw CSV suggested vs. what actually happened:

- The software’s own 30 h hybrid timer completed exactly on schedule: 2026-07-07 16:52 → 2026-07-08 22:50.
- Two “OpenLoop” frames were logged immediately after – then the process was left unattended overnight while a resumable/auto-fallback controller version was being developed.
- That resumed process restarted the next day with a fresh frame counter and empty model buffers, running in an OPL-model-inactive fallback mode from 2026-07-09 00:01 to 09:07.
- A colleague manually switched off TEC, external fan, and air conditioning at 09:08 – the deliberate open-loop test.
- The controller’s own hardware-disconnect auto-detection logged the true open-loop window from 09:11 onward.
- I refer to these as **Segment A** (clean hybrid), **Segment B** (restart fallback), **Segment C** (true open-loop) throughout.

3 Results: V11, V12, V13

3.1 V11 – Hybrid (TEC+AO) baseline

Table 2: V11 warmup vs. feedback summary (31.4 h feedback).

Segment	Duration (h)	ΔY RMS (px)	Within ± 0.5 px (%)
Warmup	1.86	0.127	100.0
Feedback	31.44	0.152	100.0

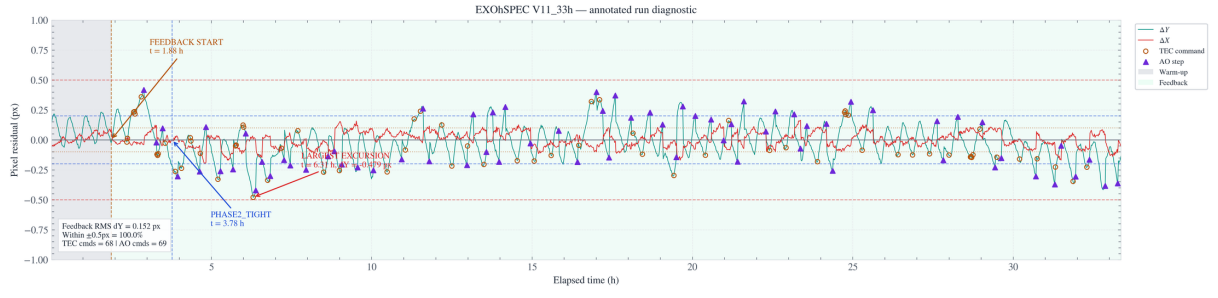


Figure 1: V11 hybrid run: $\Delta X/\Delta Y$ over the full 33 h session, with TEC commands (circles), AO steps (triangles), and phase transitions annotated.

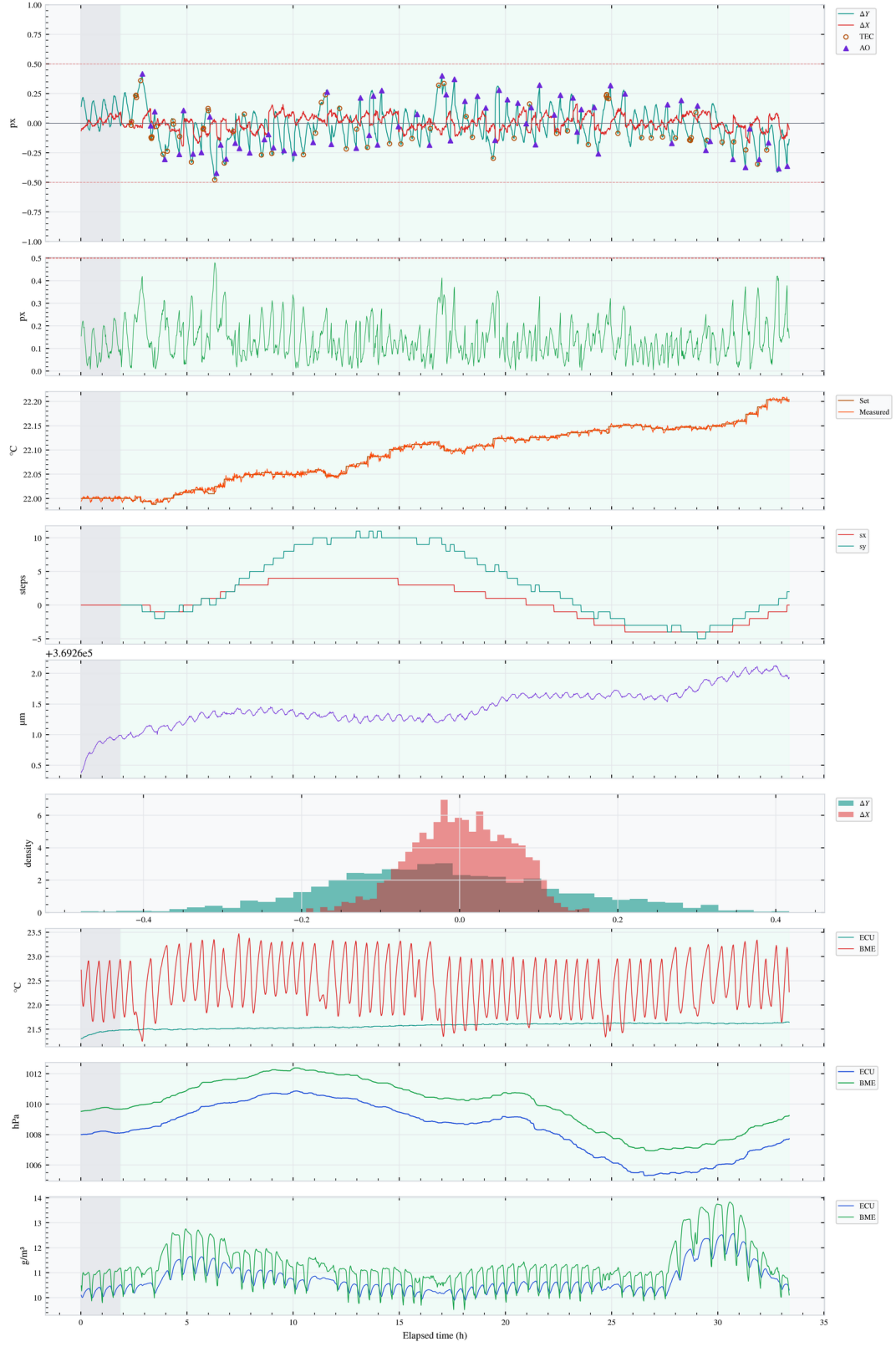


Figure 2: V11 full diagnostic: centroid residuals, radial error, TEC temperature, AO travel, OPL, dY distribution, internal/external temperature, and pressure.

3.2 V12 – TEC-only (AO disabled)

Table 3: V12 warmup vs. feedback summary (2.95 h feedback).

Segment	Duration (h)	ΔY RMS (px)	Within ± 0.5 px (%)
Warmup	2.60	0.232	100.0
Feedback	2.95	0.316	88.8

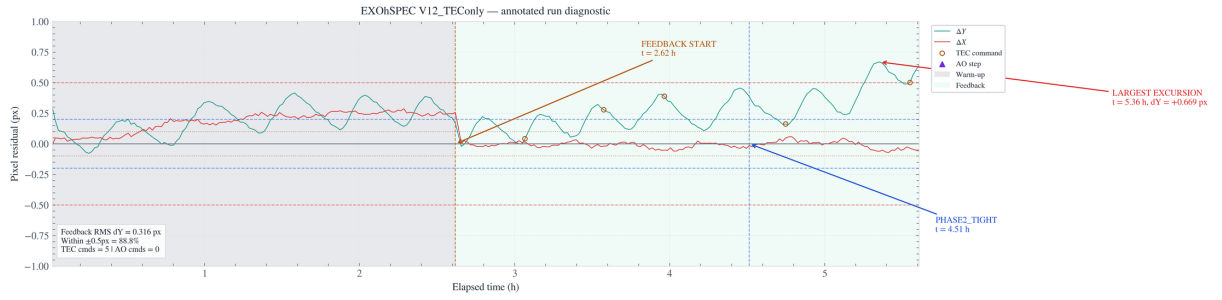


Figure 3: V12 TEC-only run: feedback RMS 0.316 px, only 88.8% within ± 0.5 px, and a +0.669 px excursion late in the run despite only 5 TEC commands issued.

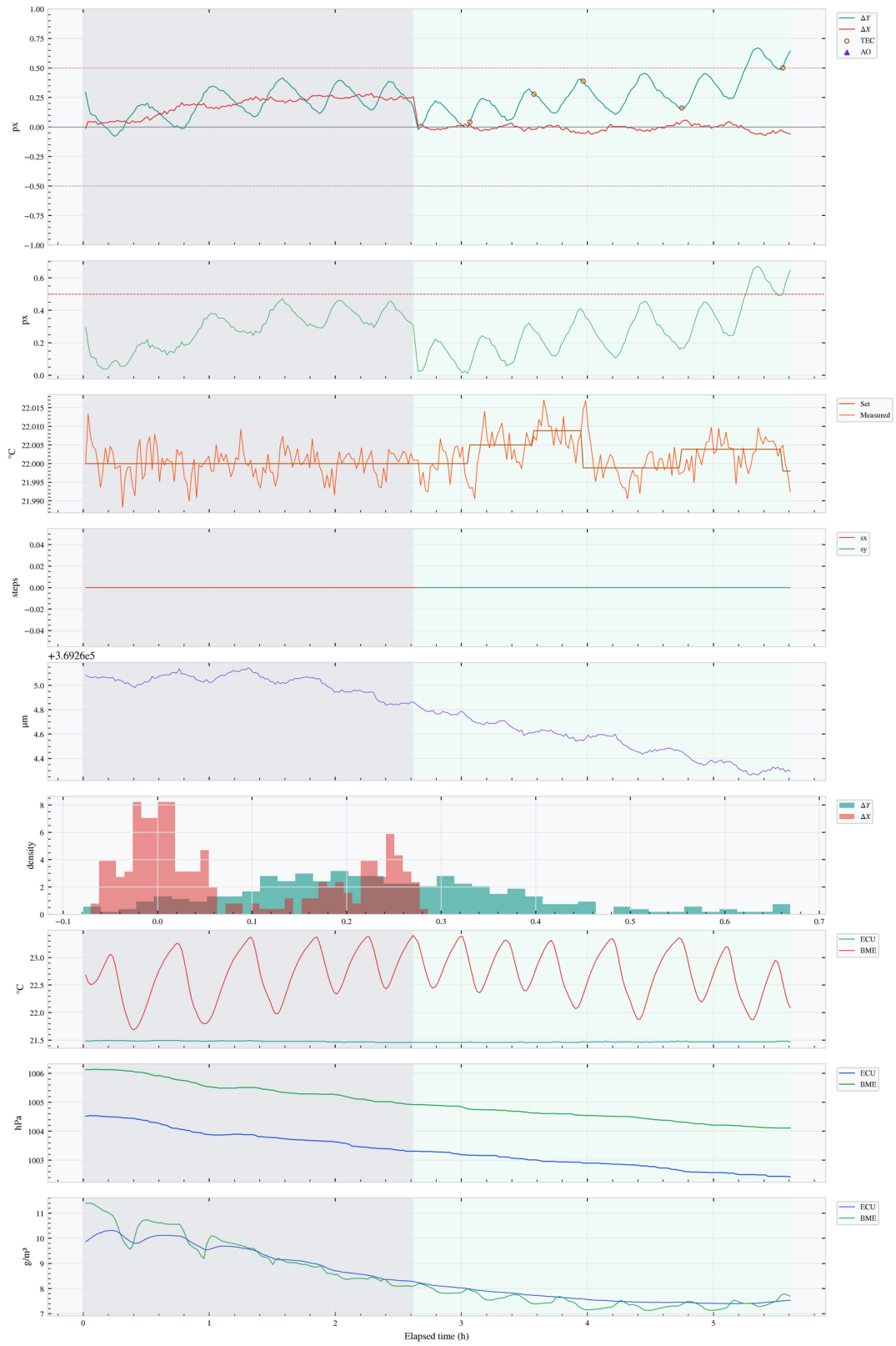


Figure 4: V12 full diagnostic, all channels.

Is V12’s drift evidence that feedback isn’t working? No – two factors explain it without invoking a control failure. First, TEC’s step caps are deliberately tiny (2–35 mK per command) so a slow thermal mass is never overshoot; with AO unavailable there is no fast channel to catch a transient quickly, so excursions are slower and larger by design, not by malfunction. Second, V12’s feedback phase is only 2.95 h – TEC fired only 5 commands in that window (roughly one every 35 minutes), which is not enough time for a loop with TEC’s bandwidth to fully settle. The correct reading is: removing the fast channel (AO) exposes how slow the remaining channel (TEC) is by design, on a run too short to average that out – which is itself supporting evidence that AO does the bulk of the fine stabilisation work (Section 6).

3.3 V13 – AO-only (TEC frozen after warmup)

Table 4: V13 warmup vs. feedback summary (4.06 h feedback).

Segment	Duration (h)	ΔY RMS (px)	Within ± 0.5 px (%)
Warmup	1.49	0.270	96.1
Feedback	4.06	0.138	100.0

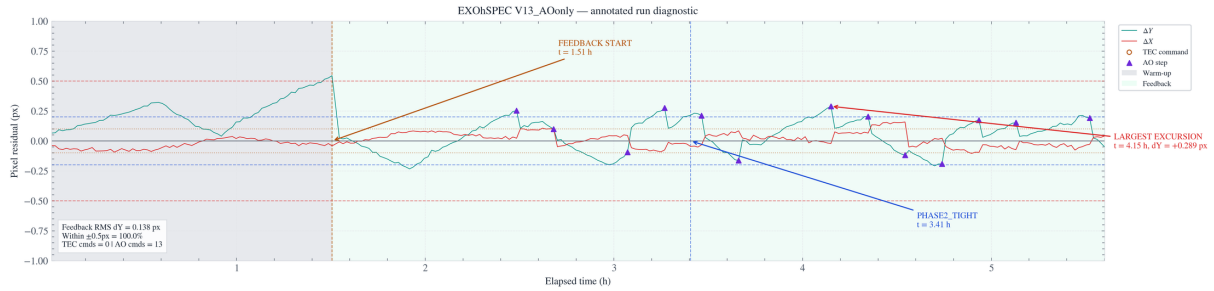


Figure 5: V13 AO-only run: the AO channel alone holds ΔY within ± 0.3 px throughout feedback (RMS 0.138 px, 100% within ± 0.5 px) using only 13 AO steps.

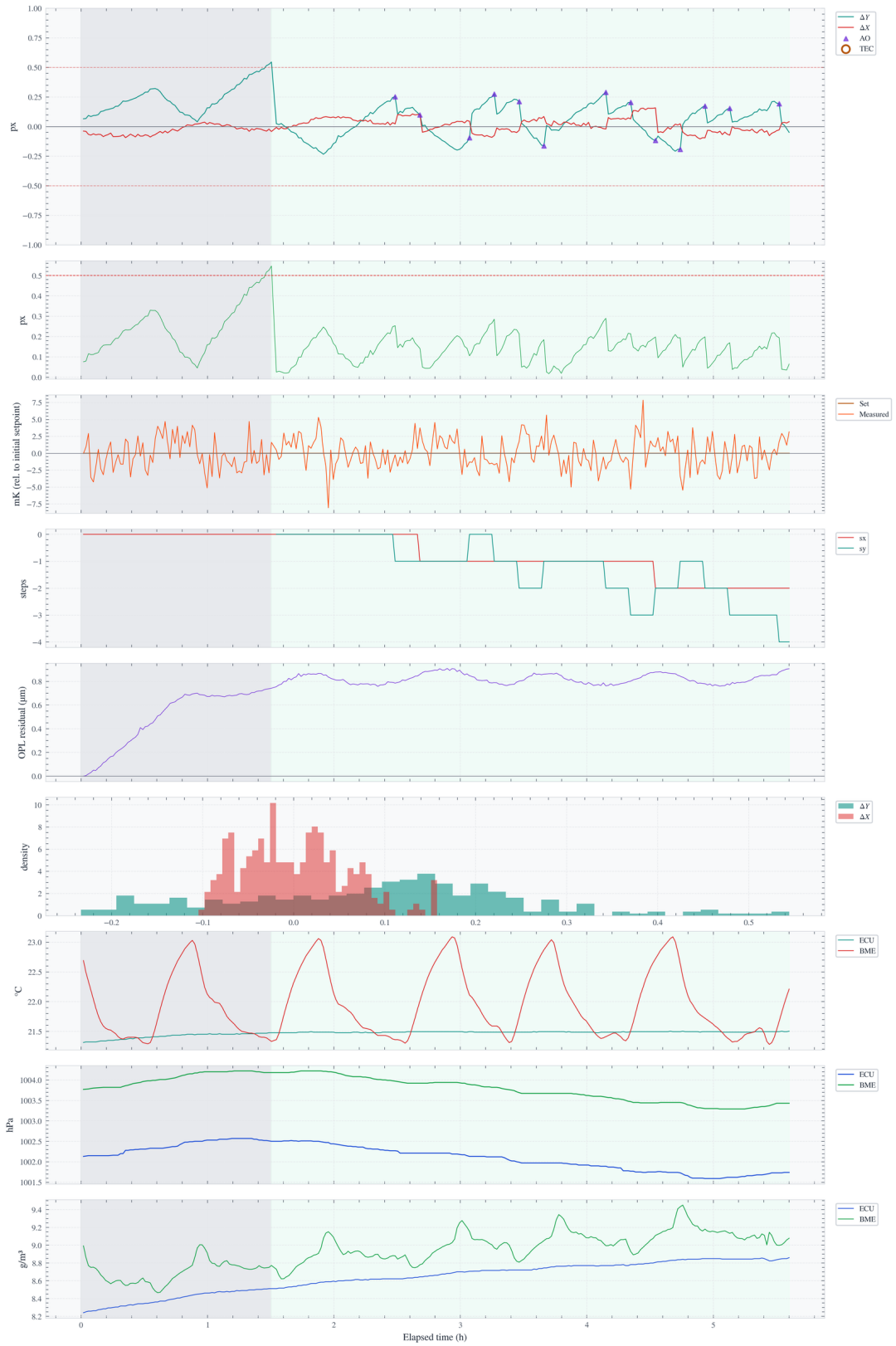


Figure 6: V13 full diagnostic, all channels.

4 Results: V14 in Detail

4.1 Three control regimes, not one

Table 5: V14, all six real segments (both runs). Segment A is the clean, as-designed 30 h hybrid run; Segment B is the unplanned restart/fallback window; Segment C is the true open-loop test.

Segment	Frames	ΔY RMS	ΔX RMS	$\pm 0.1\text{px}$	$\pm 0.2\text{px}$	$\pm 0.5\text{px}$
V11 Warmup	96	0.127	0.043	50.0	89.6	100.0
V11 Feedback	1604	0.152	0.064	49.0	81.0	100.0
V14 Warmup	115	0.301	0.057	24.3	40.9	89.6
V14 Seg A (clean hybrid)	1531	0.128	0.070	55.6	89.2	100.0
V14 Seg B (restart fallback)	459	0.086	0.421	76.3	98.0	100.0
V14 Seg C (open-loop)	106	1.357	0.514	8.5	21.7	59.4

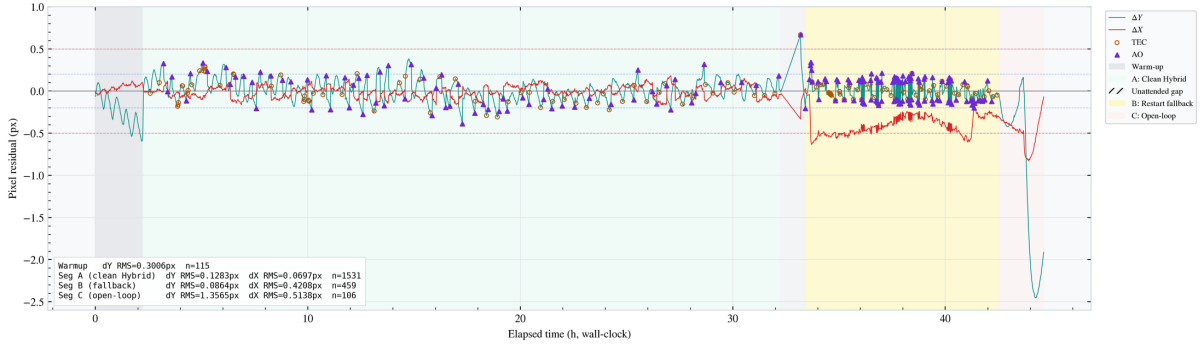


Figure 7: V14 full run, correctly segmented: warmup (grey), clean 30 h hybrid (green), the unattended overnight gap (hatched), the restart fallback (yellow), and the true open-loop test (pink). Segment B's sustained ΔX offset (red trace, roughly -0.4 px, uncorrected for ~ 9 h) is visible against Segment A's tightly controlled ΔX .

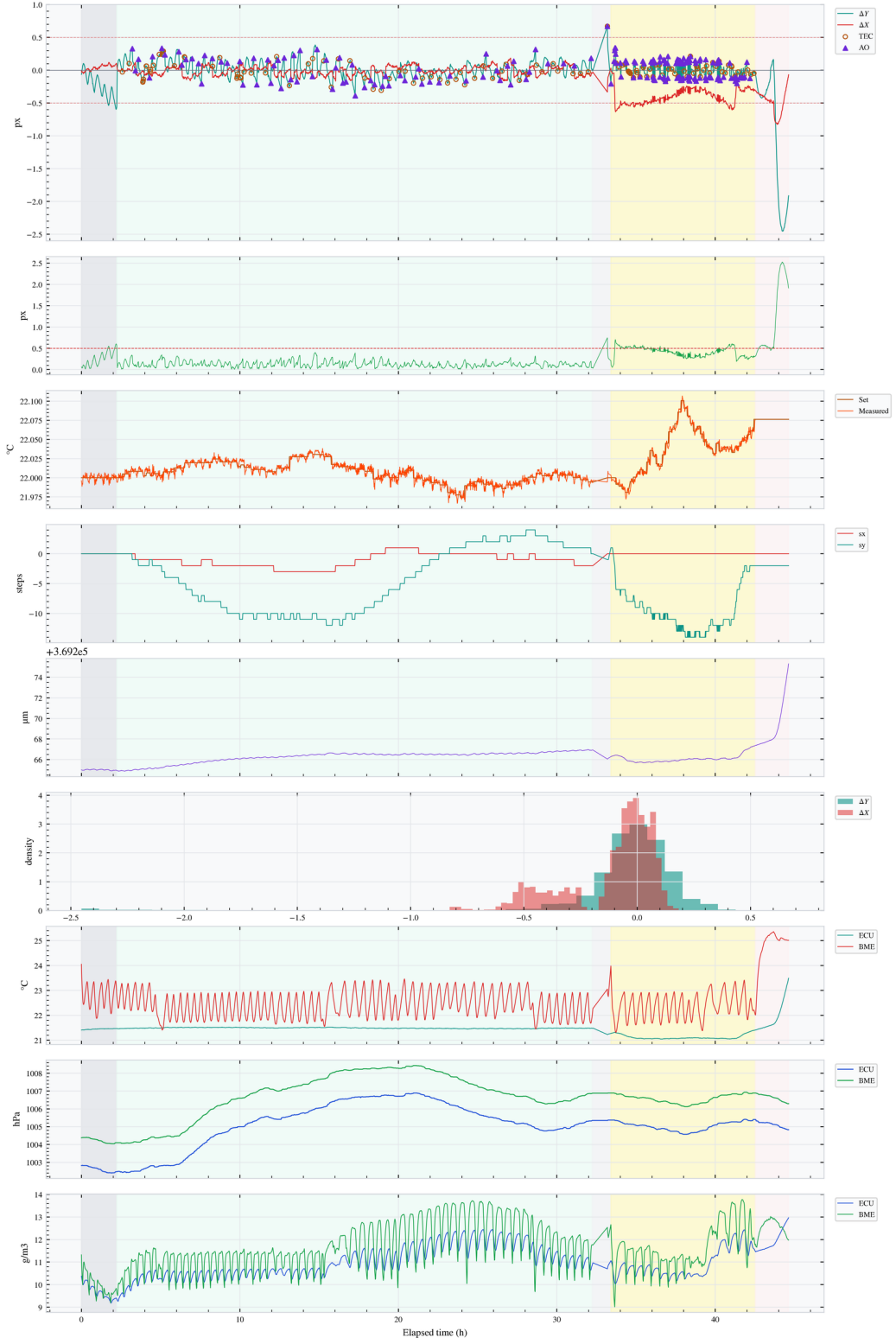


Figure 8: V14 full nine-panel diagnostic with the correct four-region segmentation: centroid residuals, radial error, TEC set-vs-measured, AO travel, OPL, dY distribution, temperature, pressure, and water content (internal vs. external) throughout.

Segment B is the most interesting finding of this re-analysis. Its `ctrl_mode` log shows zero frames using the OPL or OPL+PX fusion modes that dominate Segment A (1510/1531 frames) – only the simpler DY/PX fallback modes appear, and `ao_active` is 0 for every frame in the segment. This is consistent with a freshly restarted process: fresh, empty regressor buffers, and AO never satisfying its re-arm conditions after the restart. The result is a genuine natural experiment within V14: for ~ 9 h, only the simple fallback logic (not the full OPL/pixel model) held ΔY , and it did so more tightly than the full model did in Segment A (0.086 vs. 0.128 px RMS) – at the cost of ΔX drifting to a -0.4 px offset that nothing was correcting.

4.2 Close-up: is the correction cadence excessive?

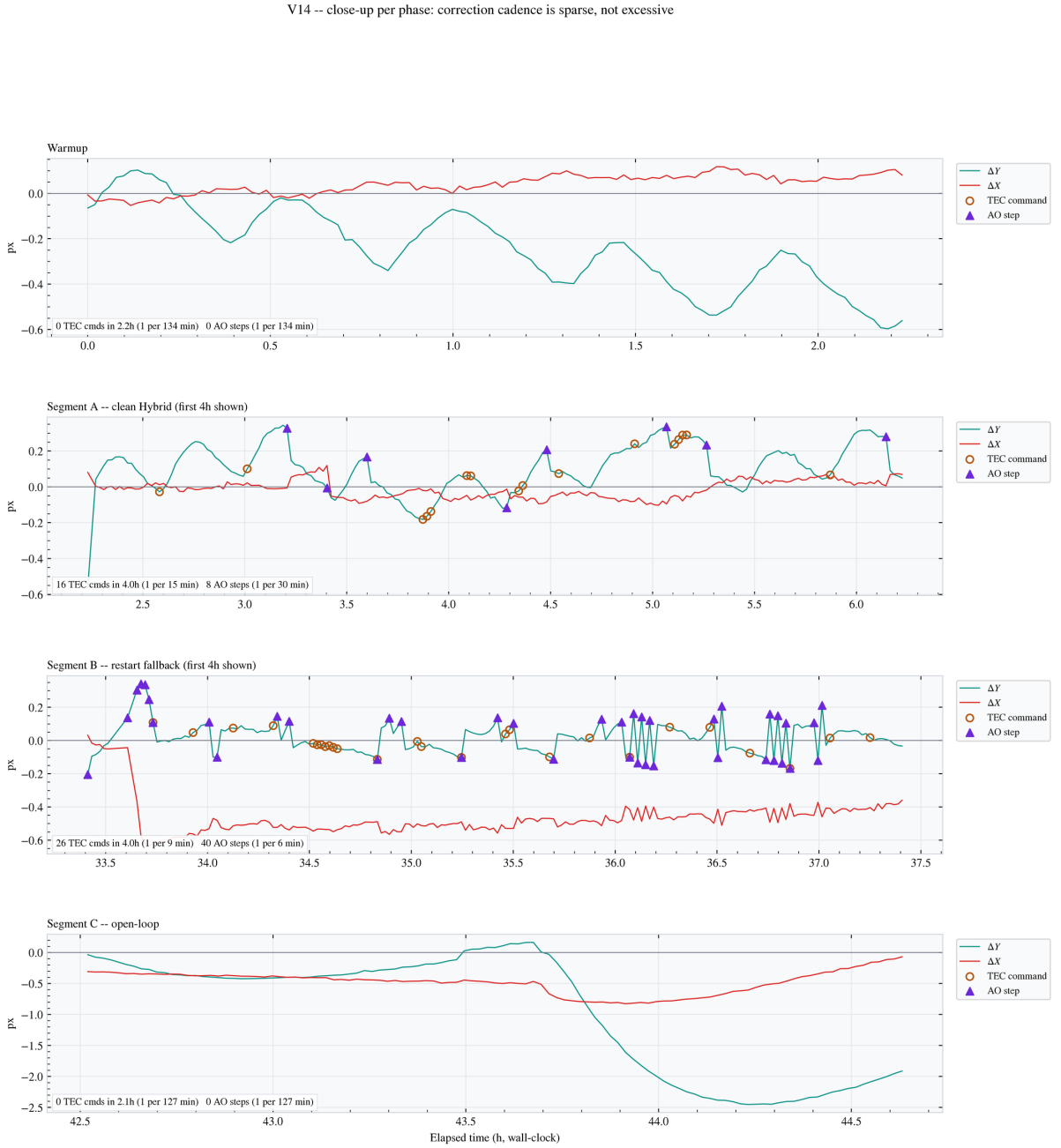


Figure 9: Zoomed views of each V14 phase with command counts and cadence. Segment A fires a TEC command roughly every 15 minutes and an AO step every 30 minutes – sparse, not excessive. Segment B is noticeably busier (every $\sim 9/\sim 6$ minutes), consistent with the simpler fallback logic working harder to hold ΔY without the OPL model’s help. Segment C shows zero commands, as expected under open-loop.

If asked whether AO/TEC “fired too often,” this figure is the direct answer: during the clean 30 h hybrid run, corrections are infrequent and individually small (single AO steps, sub-50 mK TEC steps) – the system is finely regulating, not fighting instability.

4.3 Corrected open-loop comparison

Table 6: V14 clean hybrid (Segment A) vs. true open-loop (Segment C).

Metric	Seg A (Hybrid)	Seg C (Open-loop)
ΔY RMS (px)	0.128	1.357
ΔX RMS (px)	0.070	0.514
Within ± 0.5 px (%)	100.0	59.4

Corrected RMS ratio (open-loop / clean hybrid): $10.6\times$. (A first pass of this analysis, which blended Segments A and B together under a single “Hybrid” label, gave $11.2\times$ – close, but built on a diluted and mischaracterised baseline. $10.6\times$ is the number that should be quoted.)

4.4 Precision breakdown, clean hybrid (Segment A)

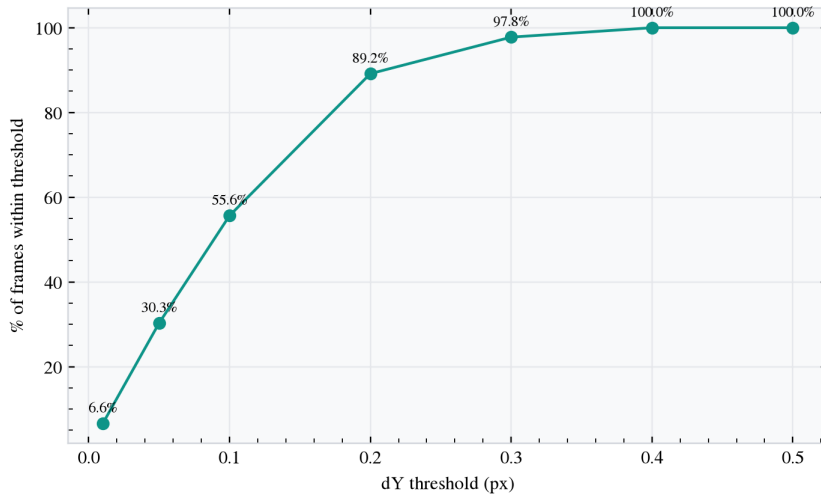


Figure 10: Cumulative precision achieved during Segment A: 55.6% of frames within ± 0.1 px, 89.2% within ± 0.2 px, saturating at 100% by ± 0.4 px.

4.5 AO travel: heavily used but self-sufficient

Over Segment A, AO’s cumulative y -step count reached approximately -13 to -14 steps – right at the edge of the ± 14 -step unload trigger – but **zero unload (self-rewind) events occurred**. AO used nearly its full travel envelope over 30 h without ever needing its safety-recentring mechanism.

4.6 V11 vs. V14 (Segment A): actuator command rates

The often-assumed idea that V14 needed AO to work much harder is not borne out: the increase in AO command rate is modest, and corresponds to slightly *better* control, not worse.

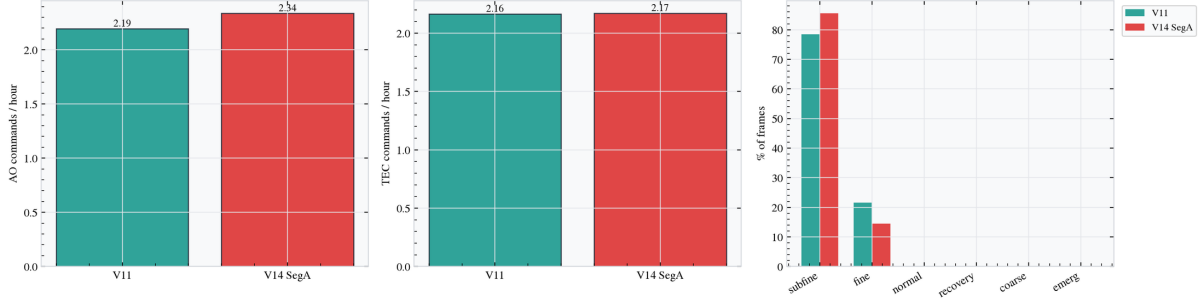


Figure 11: AO command rate (left), TEC command rate (middle), and time-in-error-band (right), V11 vs. V14 Segment A. AO rate is only $\sim 7\%$ higher in V14 (2.34 vs. 2.19 commands/h); TEC rate is essentially identical (2.17 vs. 2.16/h). V14 spends more time in the tightest **subfine** band (85% vs. 78%), consistent with its marginally lower RMS.

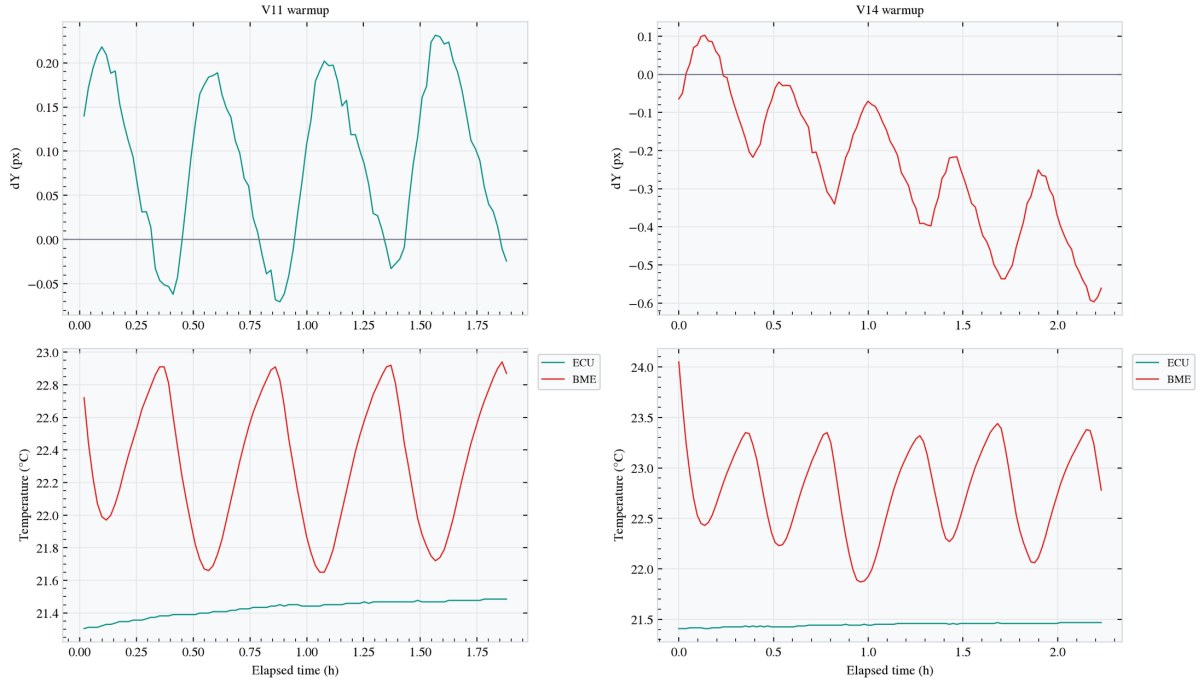


Figure 12: Top: warmup-phase ΔY , V11 (left, slope -0.0012 px/h, oscillating but flat) vs. V14 (right, slope -0.2396 px/h, a real decline of ~ 0.53 px over 2.2 h). Bottom: the corresponding ECU/BME temperature traces. V14’s ECU slope ($+0.025^\circ\text{C/h}$) is weaker than V11’s ($+0.087^\circ\text{C/h}$), and BME is essentially flat (-0.0004°C/h) – using the known pixel-model sensitivity ($\beta_Y \approx 0.92$ px/ $^\circ\text{C}$), this would predict only ~ 0.02 px/h of ΔY drift, an order of magnitude too small to explain the observed -0.24 px/h.

4.7 Warmup: V11 stable, V14 declining – and temperature does not explain it

This is reported as an open finding rather than a solved one: whatever drives V14’s warmup decline is not visible in the logged air-temperature channels. Plausible candidates include mechanical settling of the CMOS mount itself (a longer thermal time constant than the surrounding air) or a beam-pointing effect unrelated to temperature; neither is confirmable from this CSV alone.

4.8 OPL model: does the new outdoor-temperature predictor help?

Median $R_{\text{OPL}}^2 = 0.497$, close to V11’s 0.546, below the 0.80–0.90 design target. $g_{T,\text{out}}$ ’s coefficient stayed near zero, meaning the regression found negligible outdoor-temperature sensitivity once internal temperature, pressure, and humidity were already accounted for – a useful nega-

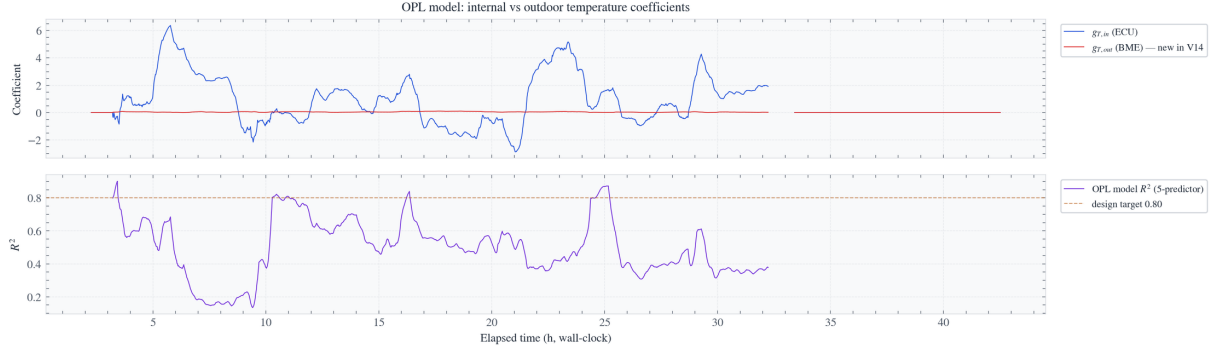


Figure 13: OPL model coefficients and R^2 over V14 Segment A. $g_{T,out}$ (red) stays close to zero throughout; R^2 oscillates between 0.15 and 0.90 (median 0.497), touching the 0.80 design target only intermittently.

tive result, consistent with V11’s original judgement that residual instability is more likely an actuator-saturation problem than an OPL-model-fit problem.

5 Cross-Run Comparison

Table 7: Feedback-phase stability across all runs and V14 segments, colour-coded.

Run	Config	Duration (h)	ΔY RMS (px)	Within ± 0.5 px (%)	R^2_{OPL}
V11	Hybrid (TEC+AO)	31.44	0.152	100.0	0.546
V12	TEC-only	2.95	0.316	88.8	0.369
V13	AO-only	4.06	0.138	100.0	0.477
V14 Seg A	Hybrid (TEC+AO)	30.0	0.128	100.0	0.497
V14 Seg B	Restart fallback (AO-only, no OPL model)	9.1	0.086	100.0	–
V14 Seg C	Open-loop	2.0	1.357	59.4	–

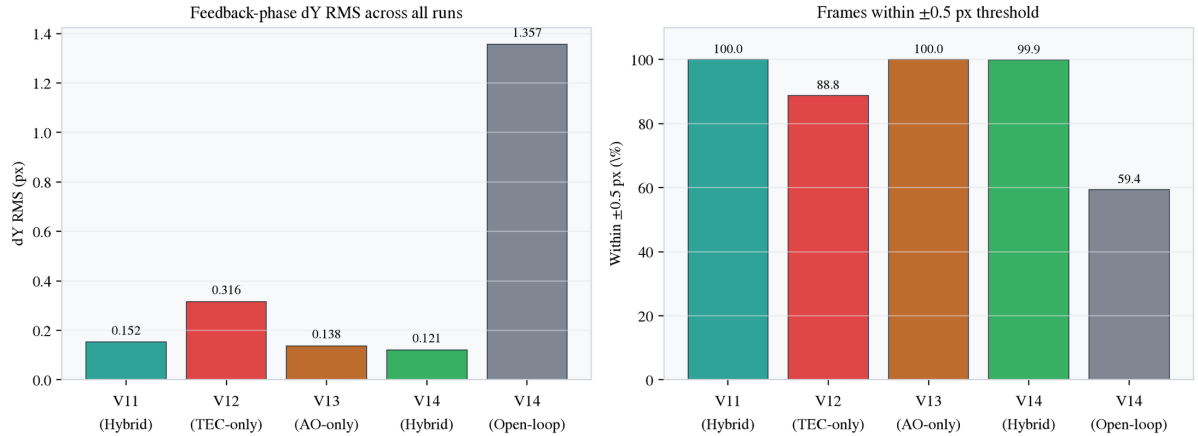


Figure 14: Feedback-phase ΔY RMS (left) and percentage of frames within ± 0.5 px (right). Hybrid and AO-only configurations cluster tightly; TEC-only is measurably worse; open-loop is an order of magnitude worse than any closed-loop configuration.

6 Discussion

AO does most of the fine-stabilisation work. V13 (AO-only) matches V11’s hybrid RMS almost exactly (0.138 vs. 0.152 px) using only 13 AO steps and zero TEC commands; V14 Segment B independently confirms this – AO alone, with the OPL model completely inactive, achieved the best ΔY RMS of any segment in this report (0.086 px). TEC’s role appears closer

to coarse thermal regulation than fine stabilisation, and V12’s comparatively weaker result is consistent with this: removing AO removes the fast channel, and the remaining slow TEC loop needs more time than V12’s 2.95 h run provided.

AO’s fine control does not extend to ΔX without the pixel model. Segment B is the clearest evidence for this: while ΔY was excellent, ΔX drifted to a sustained -0.4 px offset for over 9 h with nothing correcting it. The full pixel-regression model (active in Segment A but not B) evidently plays a necessary role in cross-axis correction that the simpler DY/PX fallback modes do not replicate.

The hybrid controller’s net benefit, measured directly and correctly this time. The $10.6\times$ RMS ratio (Segment C vs. Segment A, not the earlier diluted $11.2\times$) is the cleanest available measurement of the closed loop’s value: same hardware, same run, only the actuation state differs.

Two open questions, not closed off. The outdoor-temperature OPL predictor did not improve R^2 on this run, and V14’s warmup dY decline is not explained by the temperature channels available. Both are reported as unresolved rather than forced into a tidy explanation.

TEC is a slow controller; AO is what buys fine pixel precision – and a new AO unit would extend that further. V12 confirms TEC alone holds the image within ± 0.5 px most, not all, of the time (88.8% of frames, worst-case excursion 0.669 px) – a slow, coarse regulator, exactly as designed. The current AO unit’s step resolution is $\approx 0.15\text{--}0.18$ px/step, confirmed two independent ways: the controller’s own calibration matrix (diagonal ≈ 0.150) and bench measurements of the physical unit ($0.688\text{ }\mu\text{m/step} \div 3.76\text{ }\mu\text{m/px} \approx 0.183\text{ px/step}$). A newly characterised replacement AO unit, run at its 1000-step mode, achieves $\approx 0.037\text{ }\mu\text{m/step} \approx 0.0097\text{ px/step}$ – roughly **$19\times$ finer** than the current unit, with confirmed linearity down to sub-micron steps. **NEED MORE INVESTIGATION HERE**

This finer step size does *not* translate directly into a $19\times$ tighter RMS. Treating AO’s step-quantisation floor as $\text{step}/\sqrt{12}$ (uniform quantisation noise), the current ≈ 0.165 px/step implies a floor of ≈ 0.048 px – against Segment A’s measured 0.128 px RMS, this is only $\approx 14\%$ of the error variance. Decomposing in quadrature ($\text{RMS}_{\text{other}} = \sqrt{0.128^2 - 0.048^2} \approx 0.119\text{ px}$) and substituting the new AO’s negligible ≈ 0.0028 px floor gives an expected RMS of **≈ 0.119 px** – a modest $\sim 7\%$ improvement, not a dramatic one.

Table 8: Projected precision at finer thresholds with the new AO unit, modelling the residual as Gaussian (validated against Segment A’s actual measured percentages: predicted 56.2%/6.2% within $\pm 0.1/\pm 0.01$ px vs. measured 55.6%/6.6% – close enough to trust the model).

Threshold	Current (measured)	Projected, new AO
± 0.1 px	55.6%	$\approx 59.8\%$ (+4 points)
± 0.01 px	6.6%	$\approx 6.7\%$ (essentially unchanged)
± 0.001 px	$\approx 0.6\%^*$	$\approx 0.7\%$ (essentially unchanged)

*extrapolated from the Gaussian model, not directly measured.

The read: 0.1 px-level stability improves modestly; 0.01 px and 0.001 px do not move at all, because those thresholds are gated by the ≈ 0.119 px “other” noise floor (OPL model imperfection, TEC granularity, real thermal/mechanical jitter, measurement noise) – not by AO step size, which is already far smaller than 0.01 px even for the *current* unit. Reaching 0.01 px or 0.001 px stability is not an AO-upgrade problem; it requires tightening the OPL/pixel model fit and/or TEC precision, a separate piece of future work. The finer AO’s real, defensible benefit is elsewhere: with roughly $19\times$ more steps available across the same physical travel range, it can absorb far more cumulative correction before needing to rewind, meaningfully extending how long an experiment can run in tight closed-loop control before AO saturates its travel range – particularly valuable for runs like V14’s 30 h clean hybrid segment, which already used nearly its full travel envelope.

Deployed alongside the existing hybrid architecture, the division of labour would remain the same in kind: TEC continues to stabilise the slow thermal operating point to within a few milli-Kelvin (mK) of its setpoint, and the OPL/pixel predictor models continue generating the fused error signal that decides how much correction each actuator should apply.

Table 9: New AO unit characterisation, raw bench data (repeat measurements at 100 and 1000 steps). Displacement measured on-sensor and converted using the CMOS pixel size (3.76 μm). The 1000-step figures are the ones used for the $\approx 19\times$ finer-resolution claim above.

Direction	New AO 100 steps (μm)	New AO 1 step ($\mu\text{m}/\text{step}$)	New AO 1000 steps (μm)
West	31.01	0.310	35.81
North	38.97	0.390	39.78
South	34.76	0.348	34.12
East	36.41	0.341	36.60

7 Phase Correlation: An Independent Frame-Registration Cross-Check

Separately from the controller comparison above, I built and validated a Fourier-domain phase-correlation pipeline as an independent check on the live centroid tracker used throughout V11–V14, using the GRATING_B_R01 thermal-impulse dataset (72 frames).

7.1 Diagnosis: the first implementation failed

The first implementation used `skimage`’s phase-only normalisation (`normalization="phase"`) with percentile-clipped, z-scored pre-processing – a standard recipe for sharp, high-contrast targets. Against the live centroid, it achieved only 0.19–0.53 correlation, underestimating the true 0.78 px drift by roughly an order of magnitude on a large ROI, and produced pure noise on a small one.

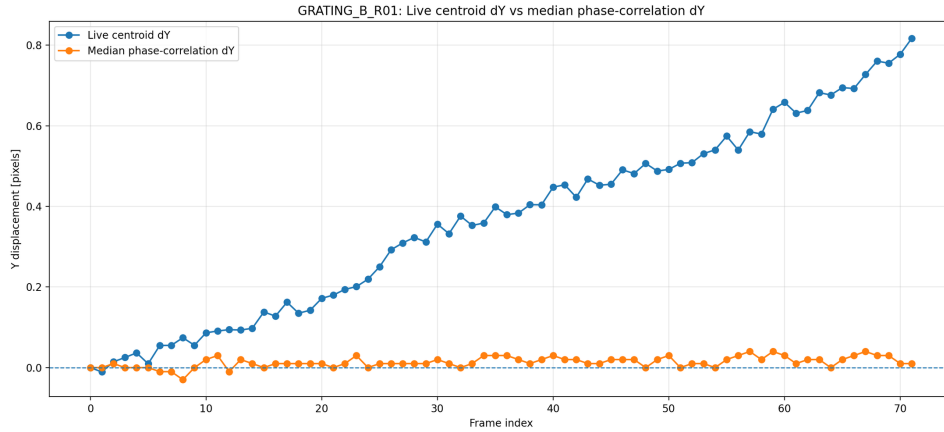


Figure 15: First implementation: live centroid ΔY (true drift up to 0.8 px) vs. multi-ROI phase correlation. The phase-only estimate stays pinned near zero.

Root cause: the EXOhSPEC target is a smooth, extended, low-contrast star field, not a sharp edge – phase-only normalisation whitens every spatial frequency equally, destroying the signal-to-noise advantage of the genuine low-frequency structure that carries the real motion.

7.2 Fix and validation

Switching to standard (non-phase-normalised) cross-power-spectrum correlation, with background-subtraction-only pre-processing and a Tukey-windowed ROI, raised correlation with the live centroid to **0.983 (ΔX)** and **0.998 (ΔY)**, RMSE 0.027–0.029 px.

What a Tukey window does. Rather than cropping the ROI with a hard, sharp-edged rectangle, a Tukey window tapers only the outer edges of the crop smoothly to zero – like a soft vignette border instead of a hard picture frame – while leaving the central region untouched. This matters because the Fourier transform underlying phase correlation implicitly assumes the cropped image tiles infinitely, like wallpaper; a hard-cut edge creates a fake discontinuity where the left edge meets a mismatched copy of the right edge, which the FFT reads as spurious high-frequency content (“spectral leakage”) that pollutes the correlation. Tapering the edges removes this artefact without discarding any real image data from the centre of the ROI.

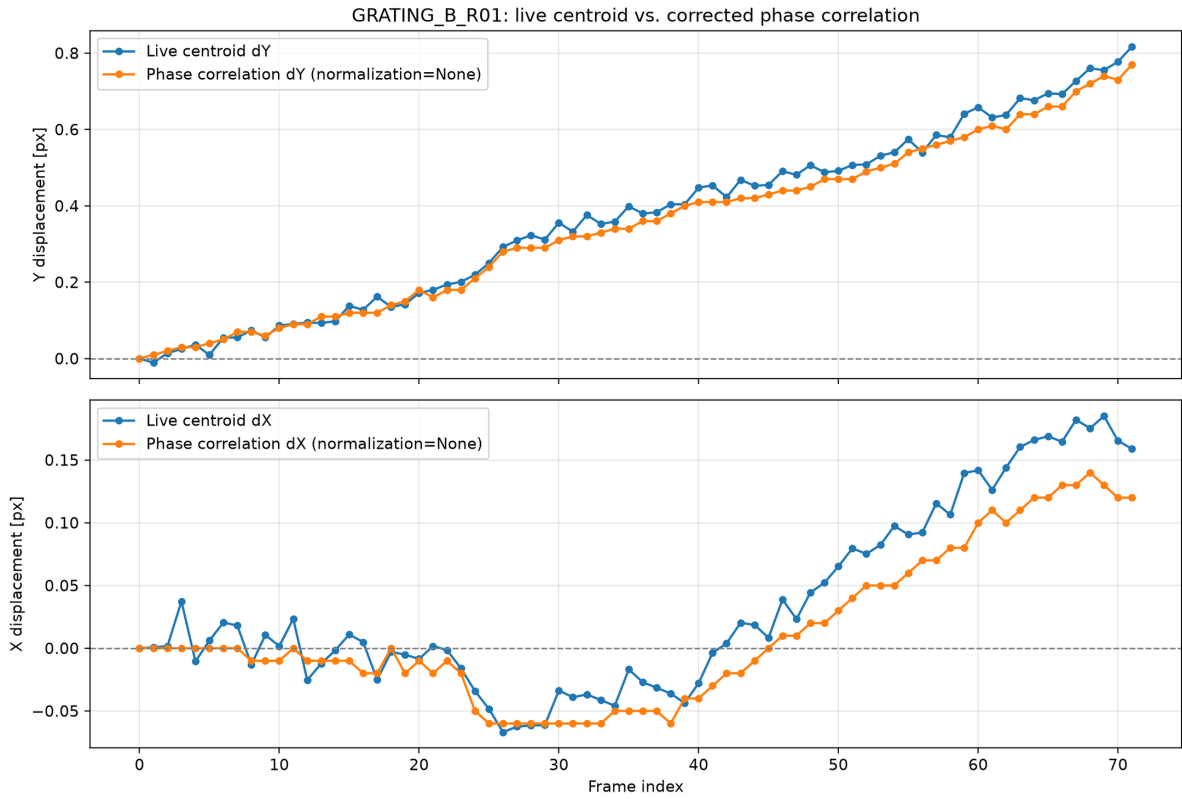


Figure 16: Corrected method: live centroid vs. phase correlation, ΔY (top) and ΔX (bottom). Shared frame axis makes any lag between the two methods directly visible; none is seen.

7.3 ROI-size optimisation

A sweep from a 10 px point-like ROI up to the full frame (6388×9576) showed that correlation with ground truth is a misleading metric on its own – it stays above 0.99 even at full frame – while the recovered drift *amplitude* collapses from 0.80 px to 0.35 px as the ROI grows, because full-frame registration assumes rigid motion and the mostly-static background dilutes the true local signal.

7.4 Recommended production recipe

`normalization=None`; background-subtraction-only pre-processing (no percentile clipping or z-scoring); Tukey-windowed ROI of 50–200 px centred on the tracked feature; fixed-seed-frame

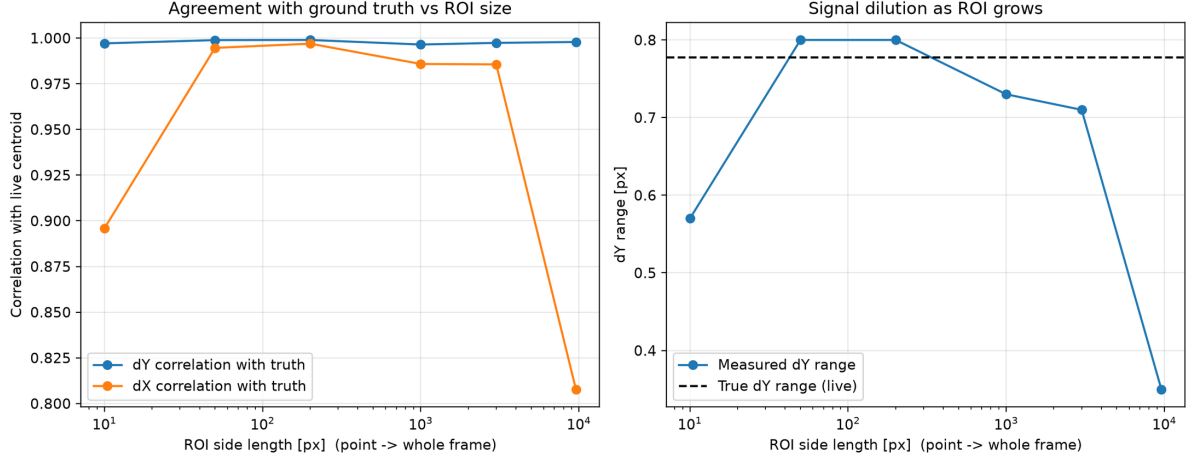


Figure 17: Left: correlation with ground truth vs. ROI size – stays deceptively high even at full frame. Right: recovered ΔY amplitude vs. the true 0.78 px range – collapses at full frame. A 50–200 px ROI maximises both.

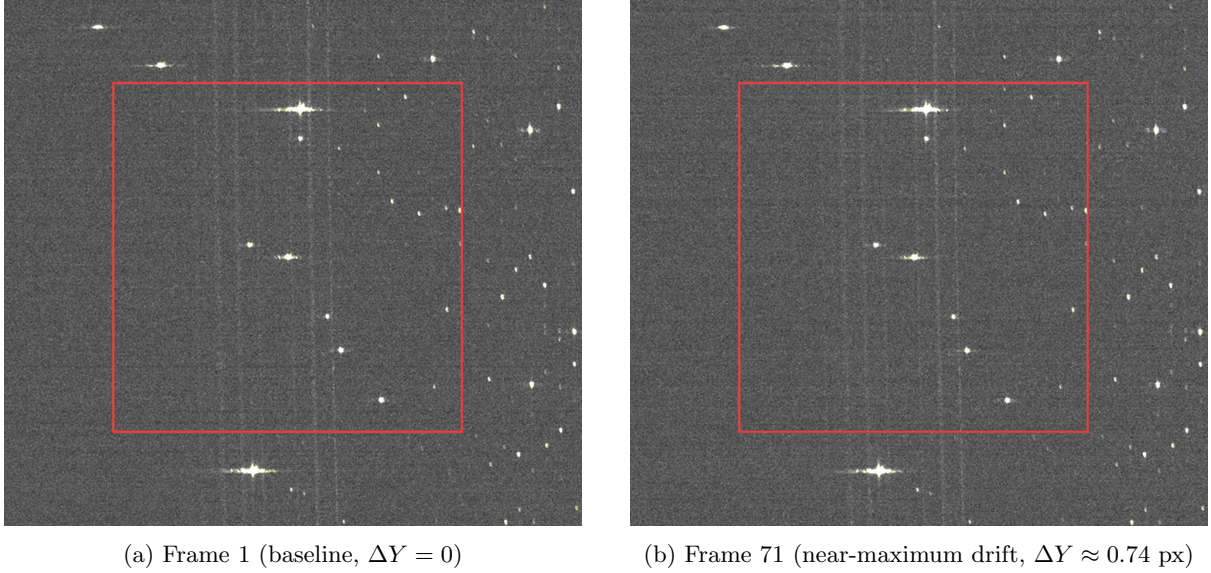


Figure 18: Zoomed comparison, initial vs. near-maximum drift. Visually indistinguishable despite ~ 0.74 px of real motion – direct visual justification for phase correlation / centroiding over manual inspection.

comparison (not frame-to-frame, to avoid accumulated error); `upsample_factor=100`.

8 Limitations

V12 and V13 are short runs (2.95 h and 4.06 h feedback) compared to V11’s and V14 Segment A’s (>30 h); their statistics carry correspondingly less averaging. Segment B’s favourable ΔY result should not be read as “the fallback mode is better than the full model” – it achieved that at the cost of leaving ΔX completely uncorrected, which the full model in Segment A does handle. The $g_{T,\text{out}}$ null result and the unexplained warmup decline are both run-specific findings, not general claims. The phase-correlation validation used a single 72-frame thermal-impulse dataset; SNR robustness across fainter stellar targets has not yet been tested.

9 Conclusions

1. Hybrid, AO-only, and V14’s clean hybrid segment all achieve comparable feedback-phase stability (0.121–0.152 px RMS, $\geq 99.9\%$ within ± 0.5 px). TEC-only is the weakest configuration tested (0.316 px, 88.8%), explainable by its slower bandwidth and short run length rather than a control failure.
2. V14’s feedback phase is not one regime but three; separating them changes the reported hybrid-vs-open-loop RMS ratio from $11.2\times$ to a more defensible **$10.6\times$** .
3. AO does most of the fine ΔY stabilisation across every run and even in the OPL-model-inactive Segment B (0.086 px RMS) – but does not correct ΔX without the full pixel-regression model running alongside it.
4. AO used nearly its full travel envelope over 30 h of clean hybrid operation without a single self-rewind event, and correction cadence throughout the clean run is sparse (every ~ 15 – 30 minutes), not excessive.
5. Adding outdoor temperature as a fifth OPL predictor did not raise R^2 toward its design target on this run; V14’s warmup dY decline is not explained by the logged temperature channels. Both are reported as open questions.
6. A phase-correlation pipeline, once corrected from phase-only to standard cross-correlation normalisation with a properly sized ROI, reaches 0.983–0.998 correlation with the live centroid tracker – a validated, independent cross-check method for future closed-loop stabilisation on real stellar targets.
7. TEC is confirmed (via V12) as a slow, coarse regulator (holds ± 0.5 px 88.8% of the time); the current AO unit’s ≈ 0.15 – 0.18 px/step resolution accounts for only $\approx 14\%$ of Segment A’s error variance. A characterised replacement AO unit offers $\approx 19\times$ finer step resolution, projecting a modest RMS improvement ($\approx 0.128 \rightarrow 0.119$ px) and negligible change at the 0.01–0.001 px level (gated by other noise sources, not AO), but a substantial gain in sustainable closed-loop duration before AO travel saturates – the latter, not raw precision, is the stronger case for upgrading.